

Certified Runtime Switching for Adversarially Robust Control

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Problem: Performance vs. Robustness

Learning-enabled controllers face an unavoidable trade-off.

- **Performance policies** excel on clean inputs but collapse under attack.
- **Defended policies** hold up under attack but pay a cost on clean inputs.

Half Cheetah	Clean	Under Attack
Performance π_{perf}	7234	3168
Defended π_{safe}	5644	5549

Goal: achieve **performance** reward on clean inputs and **defended** reward under attack.

Attacks Come In Bursts

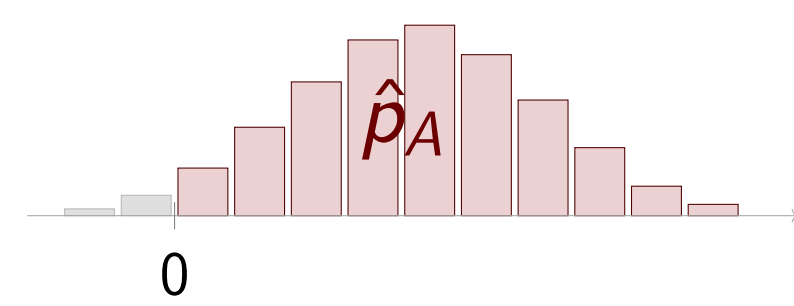
Adversarial perturbations can be **intermittent**, not constant.

- Commit to nominal $\rightarrow$ vulnerable during attack windows.
- Commit to defended $\rightarrow$ leaves performance on the table when clean.

Solution: We train a switcher classifier on adversarial perturbations to anticipate if an observation is attacked. However, we need to ensure we can trust the switcher itself.

Randomized Smoothing

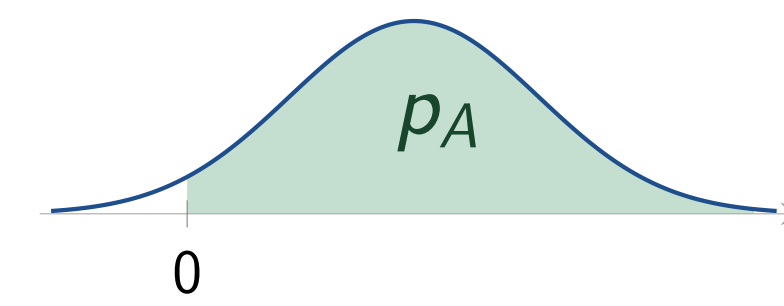
Randomized smoothing turns any classifier into a certifiably robust one. Adding Gaussian noise and taking a majority vote gives a **certified radius** $R_t = \sigma \Phi^{-1}(p_A)$.



Problem: the vote is estimated by Monte Carlo sampling. This can mean thousands of forward passes per input, making online certification infeasible.

Analytical Smoothing

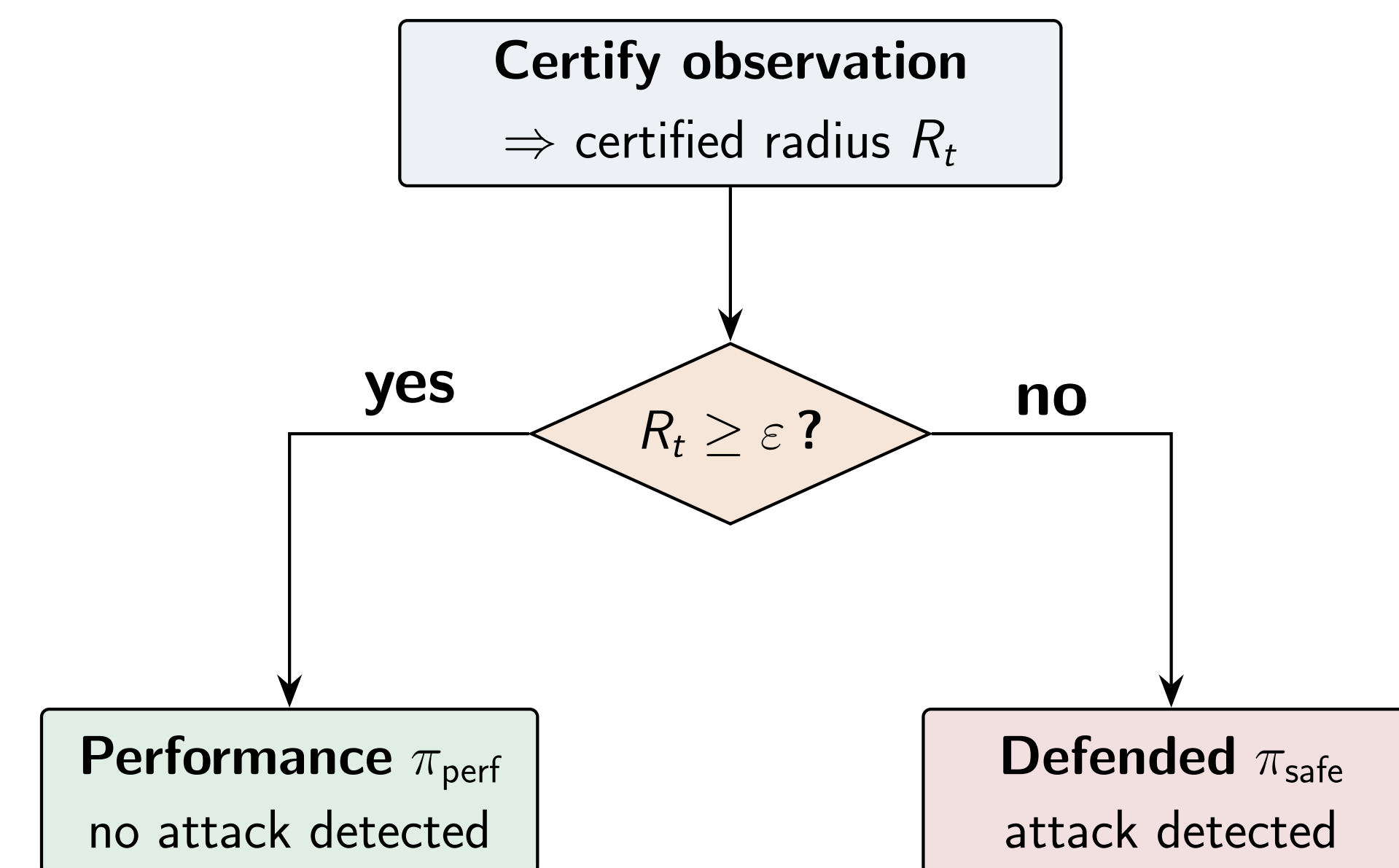
We compute the radius **analytically** by propagating the Gaussian distribution through the switching network. Using quadratic activations for the switching network, we prove the output distribution has a **known closed form**, allowing numerical integration.



R_t is computed in a **single pass** with no sampling, enabling online certification.

Certified Switcher

At each control step, the radius decides which policy is used:



If $R_t \geq \varepsilon$ (the attack budget), the observation is *provably* unaffected by an attacker $\rightarrow$ run the performance policy. Otherwise, fall back to the defended policy.

Certified Switcher Results

The certified switcher approaches performance reward on clean trajectories and better than the defended reward under intermittent attacks.

Policy	Clean	Attacked
Performance π_{perf}	7234	3186
Defended π_{safe}	5644	5549
Certified Switcher	7132	6598

Symbolic Smoothing Fits in Control Budget

Per-step certification must finish within one control interval: ~ 8 ms for Half Cheetah.

Method	Time / step
Monte Carlo (100 samples)	13.41 ms
Monte Carlo (1000 samples)	104.51 ms
Analytic (ours)	2.93 ms

Any Monte Carlo sampling certification misses the budget. The analytic method fits within the control budget, making per-step certification possible online.

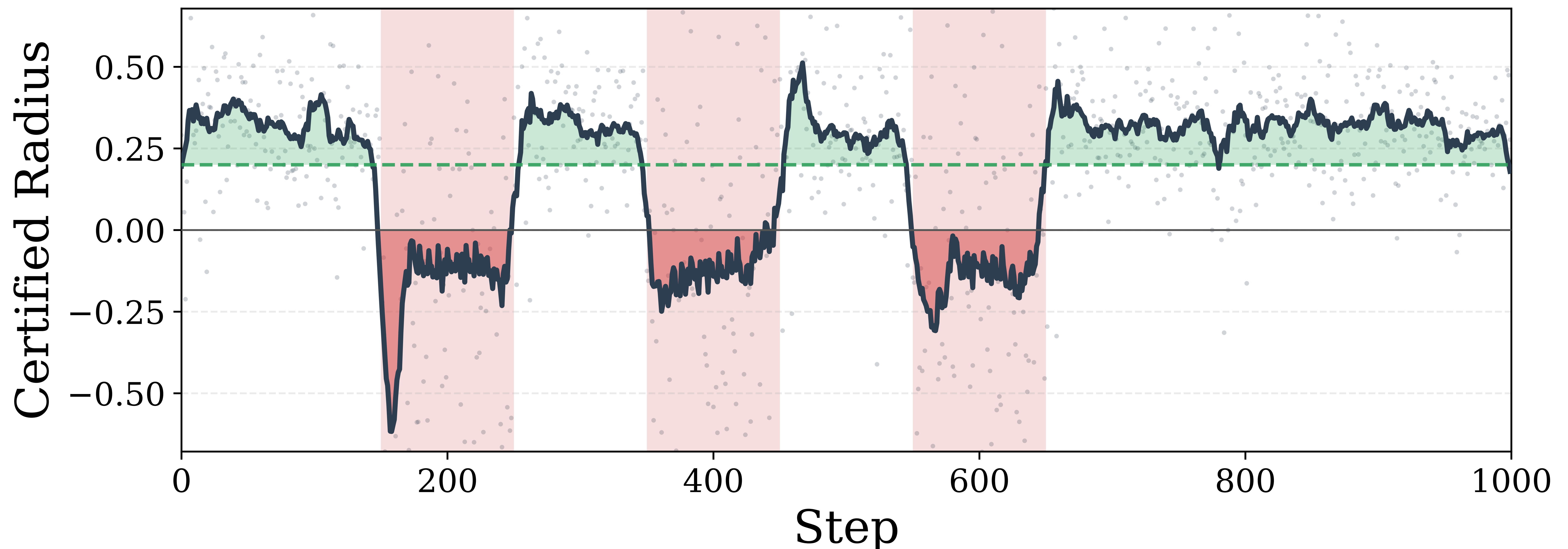
Contribution

A certified, switching controller that is provably robust per-step against adversarial perturbations and fast enough to run online, unifying analytic certification with closed-loop control.

References

- [1] Cohen et al. Certified Robustness via Randomized Smoothing. ICML 2019.
- [2] Gil-Pelaez. Note on the Inversion Theorem. Biometrika 1951.

Certified Radius over Multi-Burst Attack



R_t stays above ε (green) when clean and collapses below ε during attack windows (red). A lack of a certificate causes a switch to the defended policy.